Computer Architecture and Operating Systems
COSC 22073/BECS 22233

Lesson 02: Overview of the Computer System

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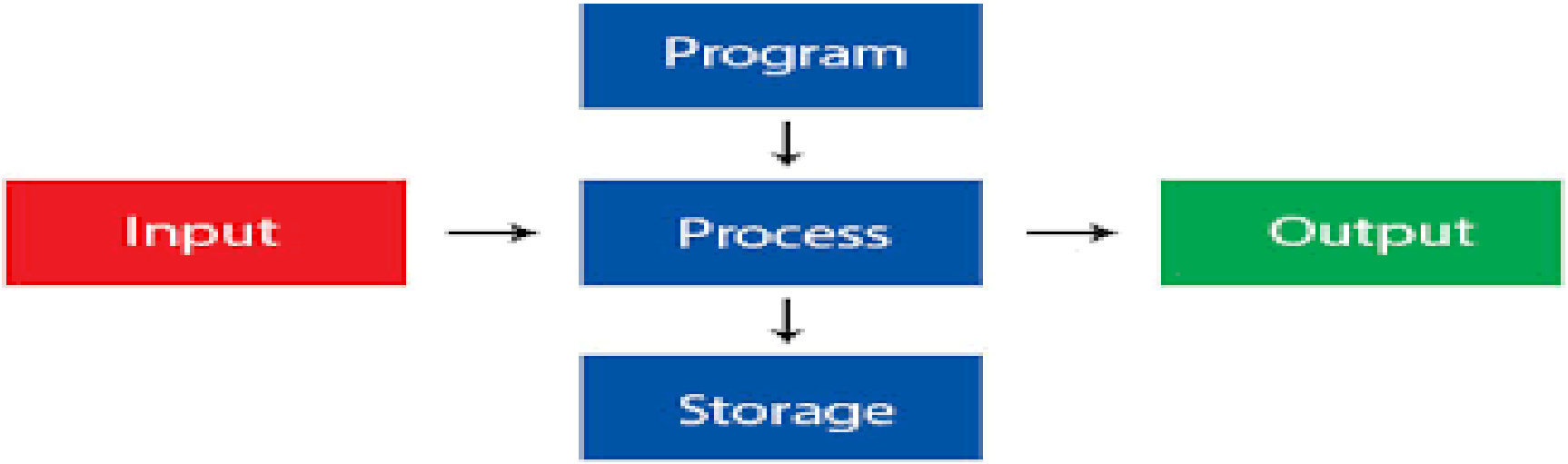
What is a Computer?

- A fast electronic calculating machine that accepts digitized input information, processes it according to a list of internally stored (memory) instructions (Program), and produces the resulting output information.



Functionalities of a computer

Any digital computer carries out five functions in gross terms:

- Takes data as input.
- Stores the data/instructions in its memory and use them when required.
- Processes the data and converts it into useful information.
- Generates the output
- Controls all the above four steps.

Classes of Computers (Based on Size)

- Supercomputers
- Mainframe Computers
- Minicomputers
- Desktop Computers
- Laptops (Notebook Computers)
- Tablets
- Ultrabooks
- Mini PCs

Classes of Computers (Based on Functionality)

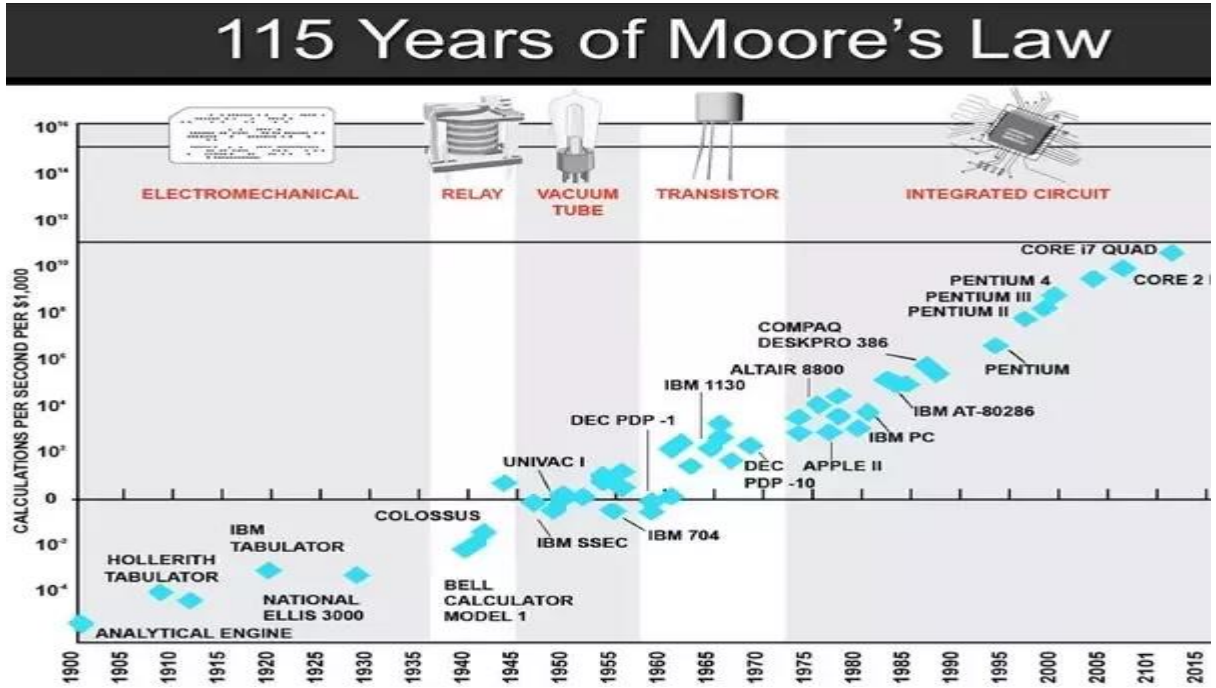
- General-Purpose Computers
- Special-Purpose Computers
- Server Computers
- Embedded Computers
- Cloud Servers and Data Centers
- Supercomputers
- Mainframe Computers
- Workstations
- Thin Clients
- Gaming Consoles
- Smartphones and Tablets
- Single-Board Computers (SBCs)
- Quantum Computers

Key features and concepts of quantum computers.

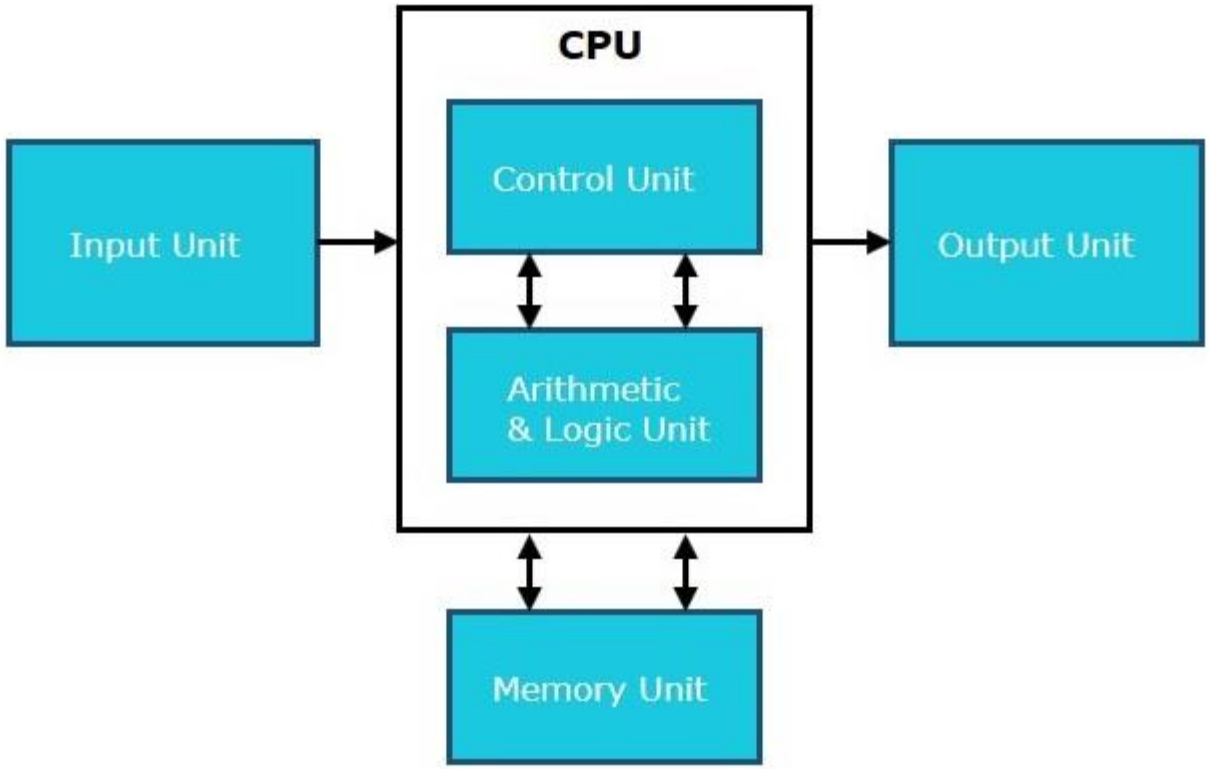
- Superposition
- Entanglement
- Quantum Interference
- Quantum Parallelism
- Quantum Tunneling
- Exponential Speedup

Moore’s Law

- In 1965, Intel’s Gordon Moore predicted that the number of transistors that can be integrated on single chip would double about every two years



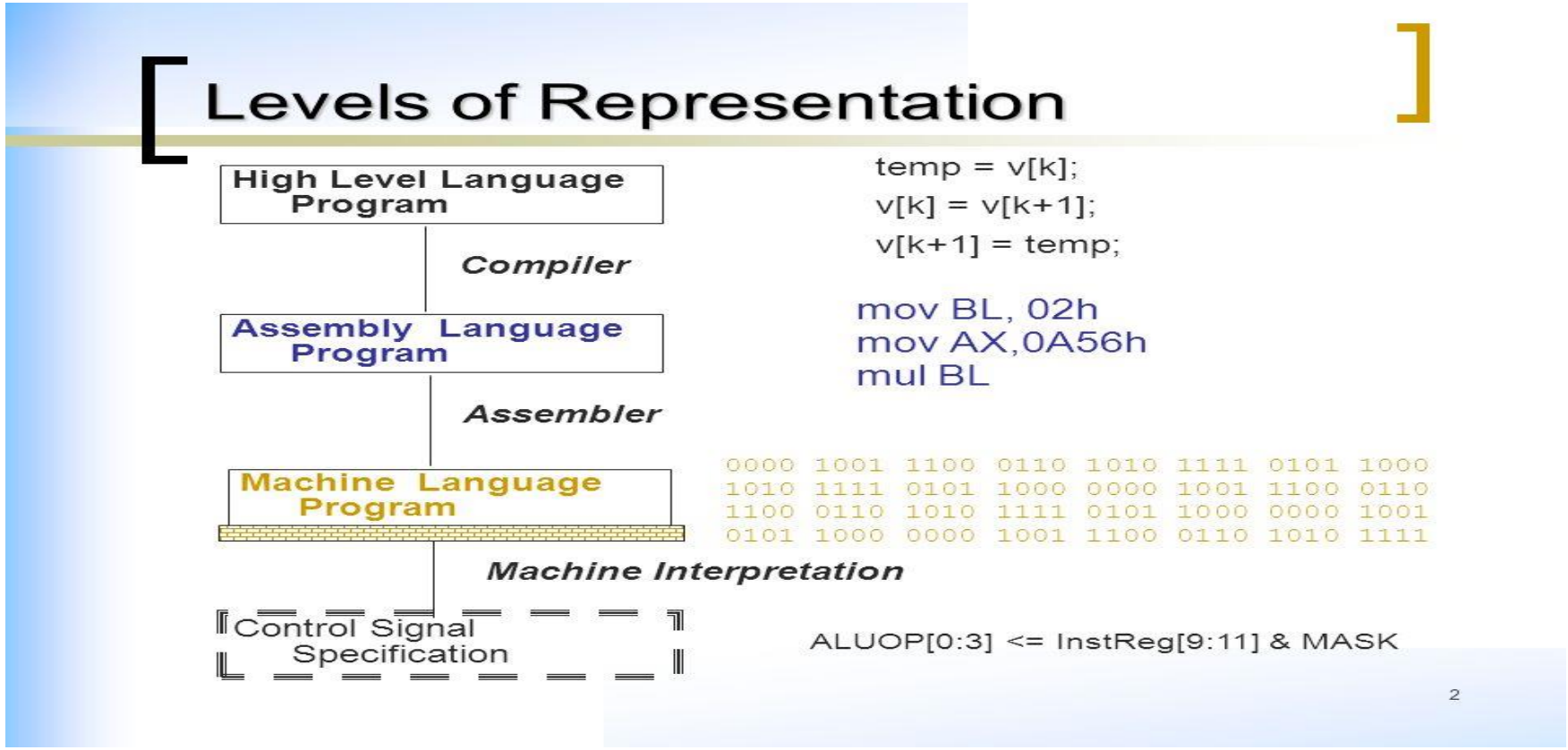
Organization of a Computer:
Four classic components of a computer – input, output, memory and processor.



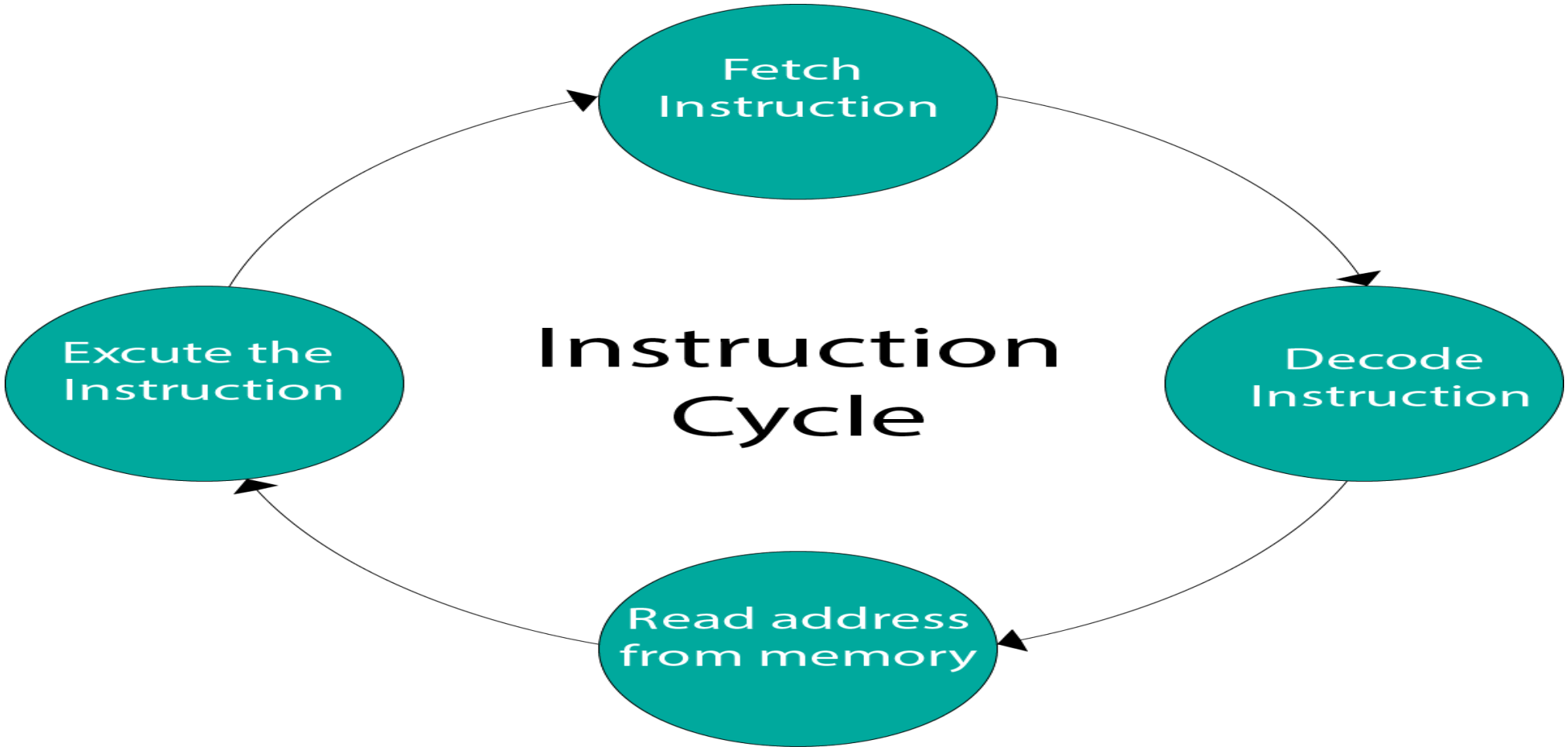
Levels of Representation

1. Physical Level
2. Digital Logic Level
3. Microarchitecture Level (Processor Level)
4. Instruction Set Architecture (ISA) Level
5. Assembly Language Level
6. High-Level Programming Language Level
7. Operating System Level
8. Application Software Level

Levels of Representation



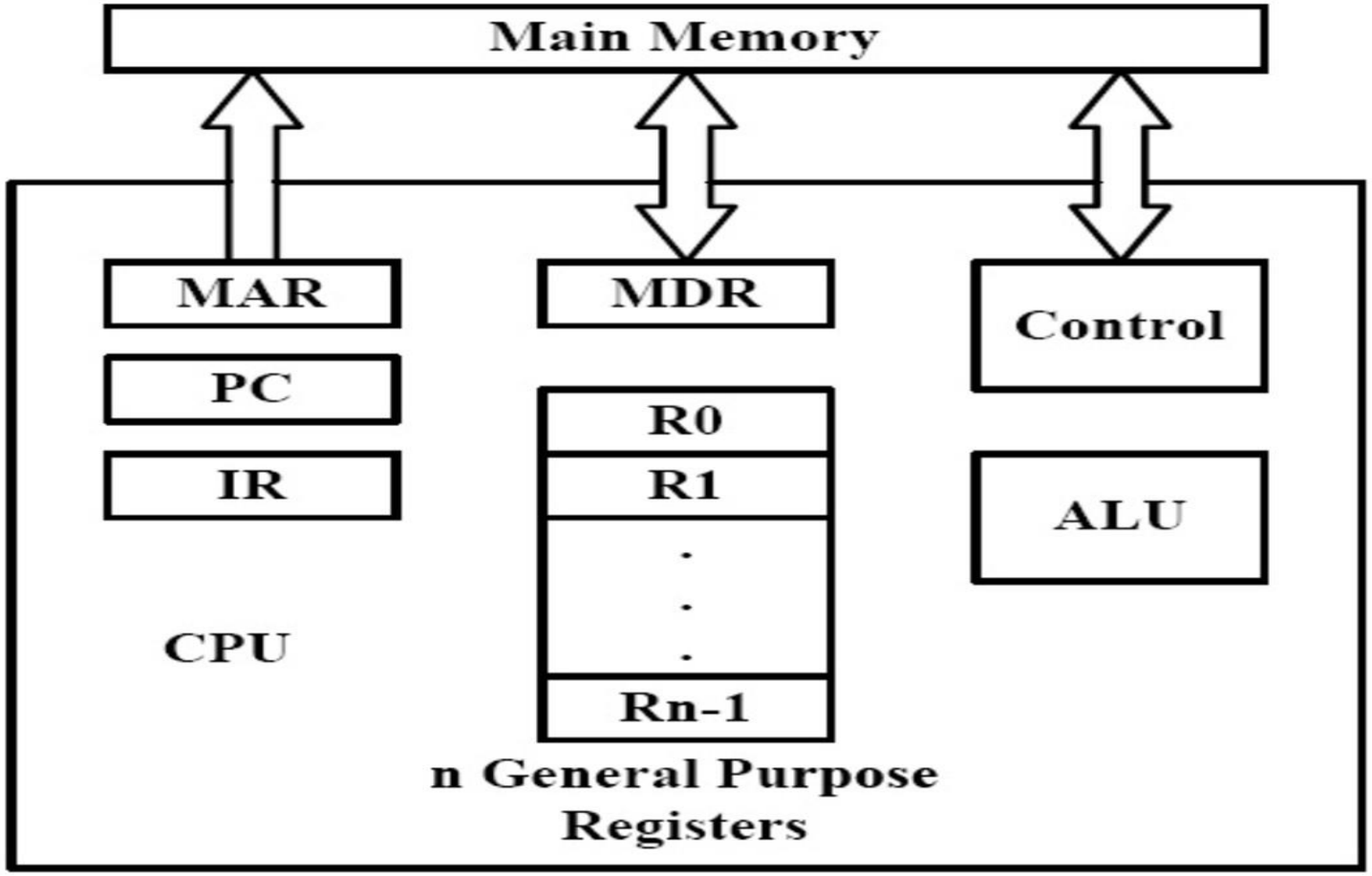
Instruction Execution Cycle



Instruction Execution Cycle

1. Obtain instruction from program storage.
2. Determine required action and instruction size.
3. Locate and obtain operand data.
4. Compute result value or status.
5. Deposit results in storage for later use.
6. Determine successor instruction.

Connection Between Processor and Memory



Exercises

1. List the steps needed to execute the machine instruction : Add R1,R2
2. List the steps needed to execute the machine instruction : Add R1,R2,R3

Thank You.